General Science

Alveary

About the Course

Learners engage with microscopes, airplanes, and sound as they practice beginning lab skills and learn to construct bar graphs from their data. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 4

About Science: Grade 4
In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

Science: Grade 4
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more “friends” in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge “by the way” and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 4
In level 4, learners begin to notice science in their community and begin to develop their own interests as they discover a variety of topics from the microscopic world, physical science, and more. A mixture of more and less challenging books is included in this course.

General Science: Grade 4
For fourth-grade students or possibly hungry third-graders or fifth-graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
4	General Science: Grade 4 1 time/week 20 min	The Elephant Scientist To Fly: The Story of the Wright Brothers The Universe in You: A Microscopic Journey

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 4				
Natural History: Grade 4		General Science: Grade 4 Nature Walks & Scouting: Grades 1-8		Labs: Grade 4 Nature Notebook: Grade 4

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 4
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 4
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

General Science: Grade 4
☐ Term 1: a museum with a microscopy exhibit
☐ Term 2: an airport observation deck and/or a museum with airplanes
☐ Term 3: a museum with a sound exhibit and a zoo

Term Prep & Teacher Tips

Science: Grade 4
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

General Science: Grade 4
☐ Term 1: introduction to rocks p.743-745
☐ Term 2: one or more birds from p.62-142
☐ Term 3: sedimentary, igneous, and metamorphic rocks p.745-748

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- lighter or a match for candle
- penny
- flashlight
- a few salt crystals
- water
- table or counter with overhang
- hair dryer
- a few sheets of multipurpose paper
- long hallway for flying paper airplanes
- large-medium-sized bowl
- paper towels
- empty can or cup
- voice recorder, such as found on most electronic devices
- friendly animal, such as yours or a neighbor's pet (Term 3)
- printables
- optional: computer
- optional: sewing needle
- optional: metal coathanger
- optional: metal spoon
- optional: 2 pieces of string ~30cm

Reminders

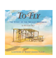
General Science: Grade 4
☐ Note that there is an activity during Lesson 1. Be sure to have your materials ready!

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

General Science: Grade 4

 The Elephant Scientist

 To Fly: The Story of the Wright Brothers

 The Universe in You: A Microscopic Journey

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
∞ View Supply List Details

General Science: Grade 4

 Hand Lens

Household Items - General Science: Grade 4

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

∞ Extra Helpings
∞ Outdoor Work

- ∞ Microscope Coloring Book
- ∞ Seek app from iNaturalist
- ∞ SkyView Lite for iOS
- ∞ SkyView Lite for Android
- ∞ Foundations (See Section 13: Science)

General Science: Grade 4

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

General Science: Grade 4

[Click THIS text or scan the QR code for links.](#)

Term 1

★ TEACHER TIP
Spend as much time as is wanted looking at the artwork in this book before going on to each successive page. Share the captioned facts, as desired.

★ TEACHER NOTE
Note that next week is an activity day. Check your supplies!

WEEK 1

⬚ **20m General Science: Grade 4 - Lesson 1**
Zooming In

> ▯ Materials: The Universe in You, penny/dime, ruler, hand lens, image link

→ INTRO
What's the smallest Thing you've ever seen? Do you think there is anything smaller? Look at the picture on the cover. What do you think you are seeing? Did you know that there are such Things that are so small that we can't usually see them? This book is about the smallest Things we know of. You can use your ruler to track how small Things are for the first few pages.

→ READ
p.1-8 "The Calliope Hummingbird" - "of a vellus hair."

→ ACTIVITY
Can you see your skin cells or bacteria? How can we know they exist? Sometimes we can sense the effect of something we can't see, like the wind; sometimes we can use a tool to make it visible. To look closer at small Things, scientists use a tool called a microscope. We do 2 things with a microscope: magnify and focus.
• Look at ∞ Image Link: Penny & Salt or Dime & Salt
• This image has been magnified. Compare it to your coin. What does it mean to be magnified? How much bigger is it? How many of your coins could fit across this image?
• Did you notice that there are some salt crystals on the magnified coin? How much has the salt been magnified? (same as the coin)
• Use your hand lens to look at your coin. What details do you notice that you couldn't see without the hand lens? Look at the other side:
Penny: Can you find the little man between the columns of the building? Notice what happens as you move the penny and lens closer together/farther apart. What does it mean to focus?
Dime: Notice that the plants on either side of the torch are different. One is an olive and the other an oak. Can you tell the difference? Notice what happens as you move the dime and lens closer together/farther apart. What does it mean to focus?

→ NARRATE & DISCUSS

WEEK 2

⬚ **20m General Science: Grade 4 - Lesson 2**
The Microscope

> ▯ Materials: video/image links, microscope, flashlight, slide, salt crystals

→ INTRO
What did we learn about in the last lesson? How do we see very small Things? What two things does a microscope do? Let's look at a very old microscope, and then we'll compare that to ours. Watch for him to show you the tube where the magnifying lenses are and the mirrors that direct the light.

→ VIEW
∞ Video Link: A 1750s Compound Microscope (5:06)

→ ACTIVITY
• Where is the tube and where are the lenses on your microscope? Refer to the diagram that came with your microscope. The one at the link may help for comparison. How much does the lens at your eyepiece (the ocular lens) magnify? How much do the lenses at the bottom of your tube (the

Term 1

	objective lenses) magnify? ∞ Image Link: Compound Microscope Parts • How did he use the mirror? Where is your light source? Place a slide with a few salt crystals on the stage. Does this direct light through the specimen or onto the surface? Look at the crystals and describe/draw what you see. • Turn the light source off and instead shine the flashlight down onto the surface of the salt crystals. Adjust the flashlight to point as much light onto the surface as you can. Look at the crystals and describe how this is different from the lower light source. **➔ NARRATE & DISCUSS**	
WEEK 3	⬚ **20m General Science: Grade 4 - Lesson 3** *Zooming in Further* ▯ Materials: The Universe in You PREP: Read Teacher Tip **➔ INTRO** What did the girl in this book see last time? Look at the pictures, as needed. **➔ READ, NARRATE, & DISCUSS** p.9-end "The smallest hairs" - "the universe within."	★ TEACHER TIP Students don't need to remember all of the scientific language. Just let them hear it "by the way."